LOWELL GENERAL HOSPITAL

KPI ANALYSIS & INSIGHTS

Analyzing Patient Safety, Staff Performance & Hospital Capacity (2020–2024)

Presented by

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Focus Areas Bed Occupancy · Staff Responsiveness · Unassisted Falls

Dataset Monthly data 2020–2024 | Source: HCAHPS / NDNQI / Internal

Tools Python · Pandas · SciPy · Matplotlib

PROBLEM STATEMENT

What operational challenge is Lowell General Hospital facing?

Mission: Lowell General Hospital is committed to putting patients first in everything we do — monitoring quality, safety, and responsiveness at all times.

The Core Issue

Fluctuating hospital capacity is creating a ripple effect on two critical outcomes:

① **Patient dissatisfaction** due to staff being stretched thin

② **Patient safety risks** from unassisted falls during high-occupancy periods

Our Objective

To investigate how hospital crowdedness (bed occupancy) impacts the nursing staff's ability to respond to patients — and whether that leads to preventable patient falls.

KPI 1 · CAPACITY

~96%

Average Licensed Bed Occupancy Rate. Captured internally. Measures the % of licensed beds filled each day.

Formula: (Patients in Beds ÷ Licensed Beds) × 100

KPI 2 · SATISFACTION

63%

Staff Responsiveness Top Box Score. From HCAHPS Survey — patients rating call-button and bathroom help as 'Always' received.

Benchmark: 65%

KPI 3 · SAFETY

2.60

Unassisted Patient Falls per 1,000 Patient Days. Captured by NDNQI — falls with no staff present to assist.

Formula: (Falls ÷ Patient Days) × 1,000

UNDERSTANDING THE KPIS

Key terms, formulas, and data sources explained

Average Licensed Bed Occupancy Rate

Source: Internal Hospital Records

Formula: (No. of Patients in Licensed Beds per Day ÷ No. of Licensed Beds) × 100

A licensed bed is a hospital bed officially approved for patient use by regulatory authorities. This KPI tells us what percentage of those beds are occupied on any given day.

Why it matters: When occupancy approaches or exceeds 95%, the hospital is operating at near-maximum capacity. This leaves little room to handle surges, puts pressure on all departments, and directly stretches nursing staff — making it harder to respond quickly to patient calls.

Staff Responsiveness Top Box Score

Source: HCAHPS Survey (post-discharge)

Formula: (Patients answering 'Always' to both questions ÷ Total respondents) × 100

HCAHPS stands for Hospital Consumer Assessment of Healthcare Providers and Systems — a standardized, publicly reported patient survey. The two Staff Responsiveness questions ask patients how quickly they got help after pressing the call button and how quickly they got help with the bathroom or bedpan.

'Top Box' means only 'Always' counts — partial responses like 'Usually' are not included. Industry benchmark is 65%. Falling below this signals patients feel staff are unavailable or delayed.

Unassisted Patient Falls per 1,000 Patient Days

Source: NDNQI (National Database of Nursing Quality Indicators)

Formula: (No. of Unassisted Falls ÷ No. of Patient Days) × 1,000

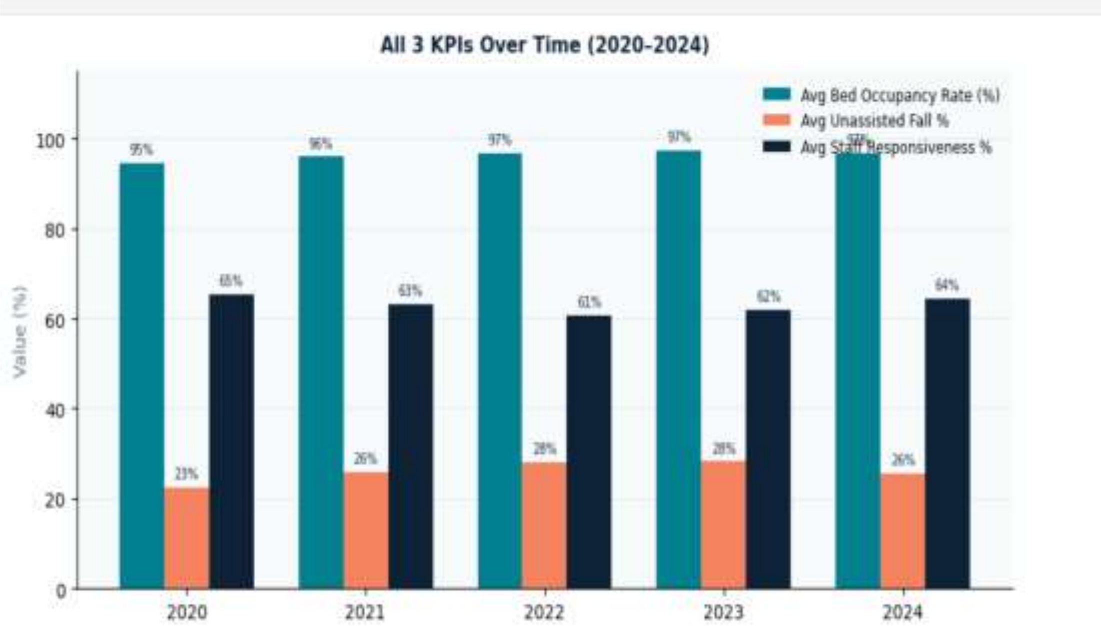
A patient fall is any unplanned descent to the floor — with or without injury. Unassisted means no staff member was present to catch or support the patient.

Patient Days normalizes fall counts by the total time patients spent in the hospital. For example, 100 patients each staying 3 days = 300 patient days. This allows fair comparison across time periods.

Why it matters: Unassisted falls are a direct patient safety signal — they are largely preventable with adequate staffing and proactive rounding.

KPI OVERVIEW: ALL THREE METRICS AT A GLANCE (2020–2024)

A side-by-side yearly comparison of all three key performance indicators



Reading This Chart

Three bars per year show how each KPI moves together or diverges:

■ **Teal** Bed Occupancy — stays high (95–97%) across all years with little variation.

■ **Orange** Unassisted Fall %— rises from 23% (2020) to 28% (2022–23) before easing slightly.

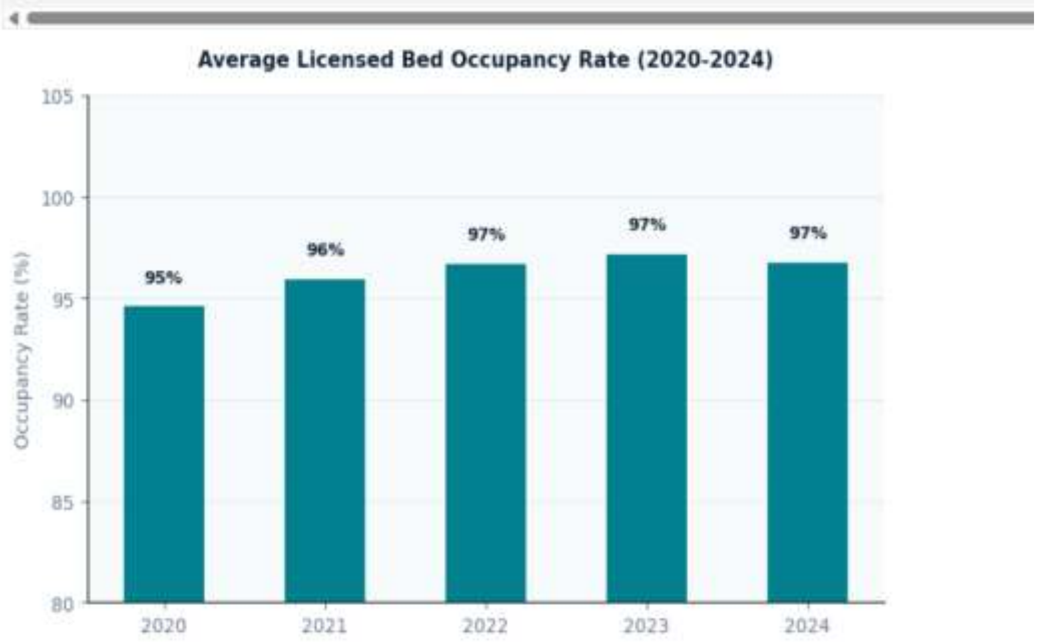
■ **Navy** Staff Responsiveness % — drops from 65% in 2020 to a low of 61% in 2022, recovering to 64% by 2024.

Key takeaway: As occupancy climbs, falls worsen and responsiveness deteriorates — in sync.

Why this chart: Before examining each KPI individually, this grouped view shows the directional relationship at a glance — all three metrics shift together, suggesting a systemic connection rather than isolated events.

METRIC 1: AVERAGE LICENSED BED OCCUPANCY RATE (2020–2024)

Is the hospital operating at safe capacity levels?



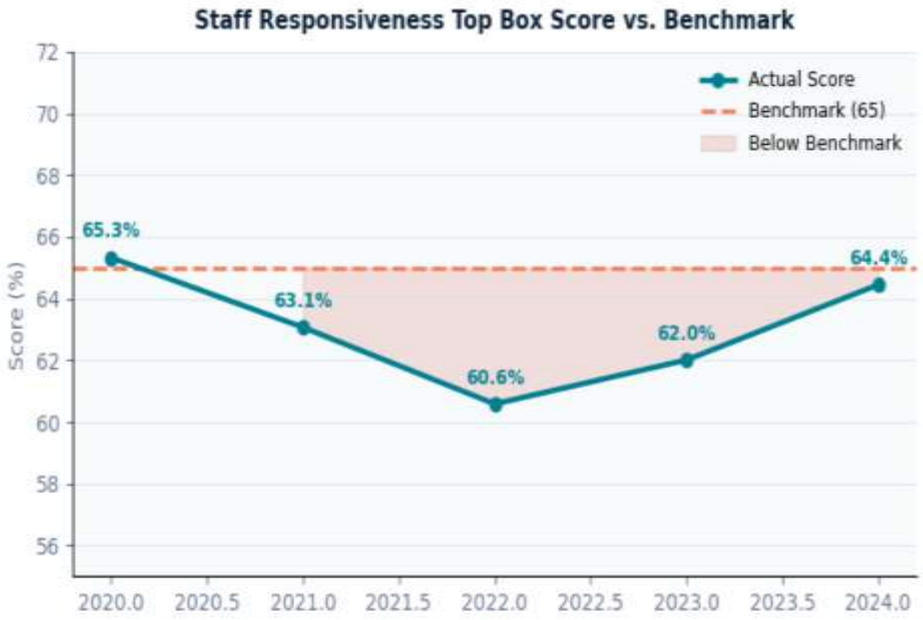
What we see: Occupancy crept from 95% (2020) → 97% (2022–2024), staying consistently at the operational ceiling.

Why 95%+ is critical: Industry best practice recommends ~85% occupancy as the safe operating zone. Above 95%, hospitals struggle to surge capacity, clean rooms quickly, or deploy staff where needed.

The diagnostic question: Does this persistent high occupancy explain the staff responsiveness decline and rise in falls we see in later charts?

Benchmark context: National average bed occupancy for acute care hospitals typically ranges 65–75%. Lowell's 95–97% is operationally strained territory.

METRIC 2: STAFF RESPONSIVENESS TOP BOX SCORE VS. BENCHMARK



65.3% 2020 — At benchmark

60.6% 2022 — Lowest point (−7% vs benchmark)

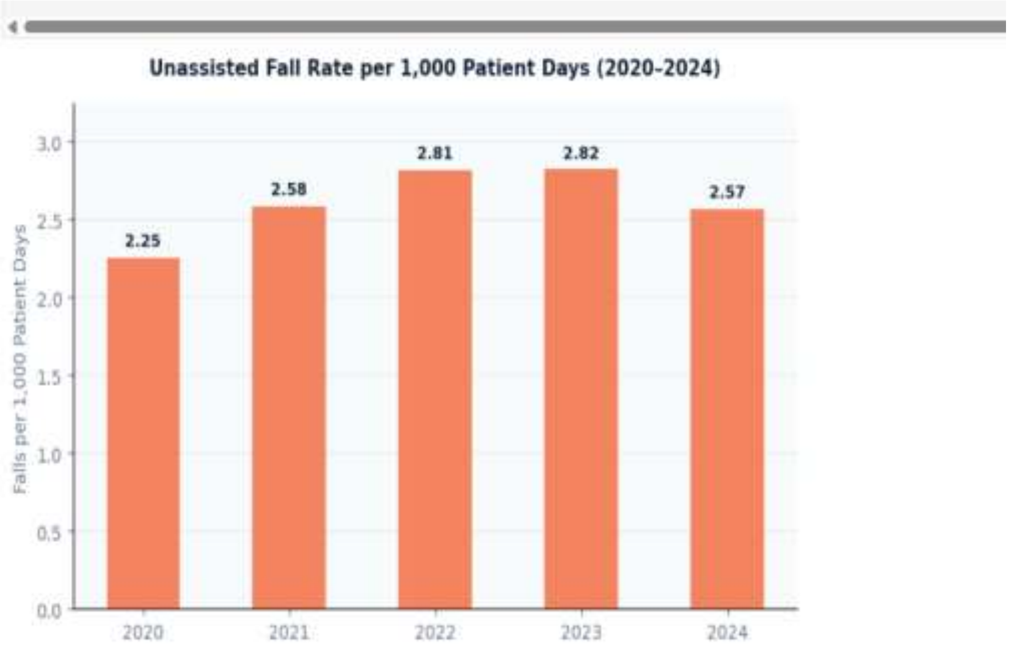
64.4% 2024 — Recovering (still below benchmark)

Benchmark: 65% is the national HCAHPS threshold. Scores below this indicate patients frequently wait longer than acceptable for call-button responses.

How to read the shaded area: The pink/red shading between the score line and the dashed benchmark line represents the 'below benchmark' zone. Wider shading = more patient dissatisfaction. The fact that Lowell crossed below in 2021 and only partially recovered by 2024 signals a structural issue, not a one-time blip.

METRIC 3: UNASSISTED PATIENT FALL RATE PER 1,000 PATIENT DAYS

Are patients falling without staff present to assist?



2020 baseline: 2.25
Starting point — relatively lower fall rate before bed occupancy reached its peak.

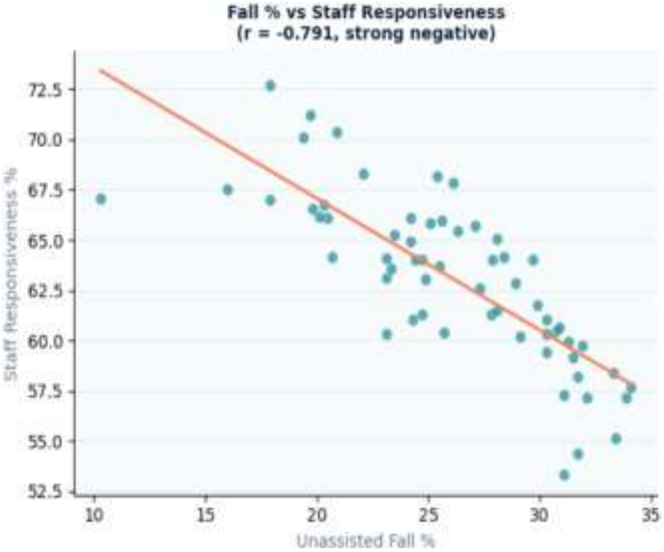
2022–23 spike: 2.81–2.82
Highest recorded fall rates — coinciding exactly with the lowest staff responsiveness scores. Staff stretched too thin to prevent falls.

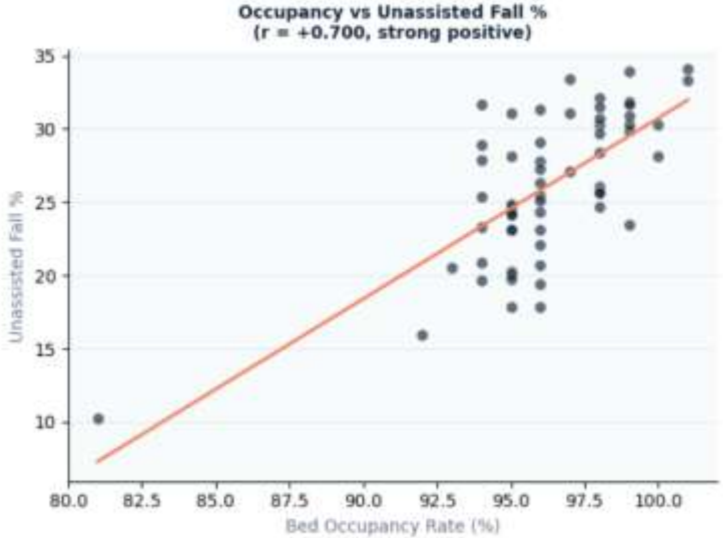
2024 improvement: 2.57
A slight recovery mirrors the partial rebound in staff responsiveness. Still above baseline — sustained action needed.

The 'per 1,000 patient days' unit normalizes for volume — it accounts for how many patients were in hospital and for how long, so comparisons across years are fair even if total patient volume changed.

STATISTICAL ANALYSIS: HOW DO THE KPIS RELATE TO EACH OTHER?

Pearson correlation coefficients reveal the strength and direction of relationships





r = −0.791 | Strong Negative Correlation

When unassisted falls increase, staff responsiveness drops sharply. Staff overwhelmed = patients falling unattended. Each 1% rise in fall rate coincides with a measurable dip in responsiveness score.

r = +0.700 | Strong Positive Correlation

Higher bed occupancy directly drives more unassisted falls. As the hospital fills up, nursing staff are spread too thin — patients attempting to move independently without waiting for help.

How to read scatter plots: Each dot is one month of data. The orange diagonal line is the linear regression (trend line). Dots clustering tightly around this line indicate a strong relationship. The Pearson r value (−1 to +1) quantifies strength: |r| > 0.7 is considered a strong correlation.

SEASONAL TRENDS: FALL RATE & STAFF RESPONSIVENESS BY MONTH

Are there predictable times of year when the hospital is most at risk?

How to read: Orange bars (left axis) = average fall rate. Teal line (right axis) = average staff responsiveness score. When bars are tall and the line dips simultaneously, it confirms the inverse relationship is not just yearly — it repeats seasonally.

Seasonal Patterns

High-risk months:
Jan, Feb, Sep, Dec show elevated fall rates alongside lower staff responsiveness — a consistent inverse pattern across years.

Lower-risk window:
Mar–May and Jul–Aug see relatively better responsiveness scores with moderate fall rates.

Why winter & year-end?
Flu season drives higher admissions. Holiday staffing shortfalls reduce nurse availability. Year-end budget constraints may limit float pool access.

JAN	SEP	APR	DEC
High risk	Lowest resp.	Best resp.	High risk

SOLUTIONS & RECOMMENDATIONS

Data-driven actions for nursing staff and hospital leadership

01 Dynamic Staffing Models

When bed occupancy is forecasted to exceed 95%, automatically trigger additional support staff or float pool nurses specifically assigned to call-button coverage.

Data rationale: r = +0.700 between occupancy and falls confirms that overfill = under-support. Pre-emptive staffing breaks this chain.

02 Proactive Bathroom Rounding

During peak risk months (Jan, Sep, Dec) and during high-occupancy shifts, mandate scheduled nursing checks every 2 hours — proactively assisting patients before they attempt to move unassisted.

Data rationale: Monthly seasonal analysis shows fall rate spikes are predictable — they can be anticipated and staffed for, not just reacted to.

03 Targeted Staff Training

Train nursing staff on prioritization protocols during high-occupancy shifts — specifically how to triage call-button responses and bathroom requests to minimize time-to-response.

Data rationale: r = −0.791 between responsiveness and falls is the strongest signal in this dataset. Improving responsiveness even slightly during peak periods could yield significant fall reduction.

SUMMARY

Objective: Analyze how hospital capacity affects staff responsiveness and patient safety across 2020–2024 using three interconnected KPIs.

95–97%

Bed Occupancy

Consistently at or above operational ceiling — no buffer for surges. National average is 65–75%.

60.6%

Staff Responsiveness

Hit a critical low in 2022 — 7% below the 65% HCAHPS benchmark. Partial recovery to 64.4% by 2024.

2.82 peak

Unassisted Falls

Worst in 2022–23, aligning precisely with the lowest responsiveness period. Improved slightly in 2024.

–0.791

Correlation (r)

Strongest signal: responsiveness and falls are tightly linked. When one drops, the other spikes — predictably.

Conclusion: This analysis confirms that bed occupancy, staff responsiveness, and patient falls are not independent events — they form a chain reaction. Reducing falls requires breaking this chain upstream: through predictive staffing, proactive rounding protocols, and targeted training during high-risk periods. These interventions are low-cost, data-backed, and implementable without capital investment.

THANK YOU

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Open to questions and discussion